Sparse FGLM for the linear stage of `jv_cover`: C_{cov} 1.27–1.31x cheaper, S 1.28x cheaper, same relations

2026-10-19. Run report: `reports/run-20261018T2138Z/report.pdf`. Class: **engineering** (same algorithm, cheaper linear algebra).

Question

After the `f4_fp` speed-up, the linear stage after F4 is 40 % of one Nagao test (4.8e5 of 1.22e6 mults at p = 101). The derived estimate was 210k–260k for it and C_{cov} 1.22–1.29x cheaper. Does an exact sparse path beat that?

What was built (`solve_zero_dim_mode`, `src/cryptanalysis/jv_cover.rs`)

- Only M_{x_6} is built. Its 44 unit columns are shifts (a target index); the 20 others are stored as sparse vectors.
- Minimal polynomial by Wiedemann: $s_i = u^T M^i v$, $i < 2\delta$, Berlekamp–Massey; up to 4 draws. Degree δ means minimal polynomial = characteristic polynomial, so no root is missed and x_6 separates the points. Else: dense fallback (both costs counted, `lin_fallback`).
- Roots by `PolyRing::roots`. For each root r the eigenvector w of M^T satisfies $w[x_6 b] = r w[b]$, so it is fixed by one unknown per shift chain (20); the dense columns give a 20x20 system of corank one; $x_i = \frac{w[x_i]}{w[1]}$, $x_6 = r$. Corank not one, or a missing degree-1 standard monomial: dense fallback.
- Every candidate is still checked against all input polynomials.
- `JV_LIN=dense|l1a|sparse` (default sparse) selects the arm in one binary. L1a = dense Hessenberg charpoly of M_{x_6} + the same chain recovery.
- All arms draw one `u64` from the caller's rng and use a private rng after that, so downstream randomness is identical across arms. This changes the stream relative to older main (baseline numbers move by under 0.1 %).
- Unit test `sparse_linear_stage_agrees_with_dense` (p = 53, 101; 4–6 variables; 48 systems, planted and random): dense, sparse and L1a return identical solution sets; passes in release.

One decomposition test (ccov, 120 tests per prime)

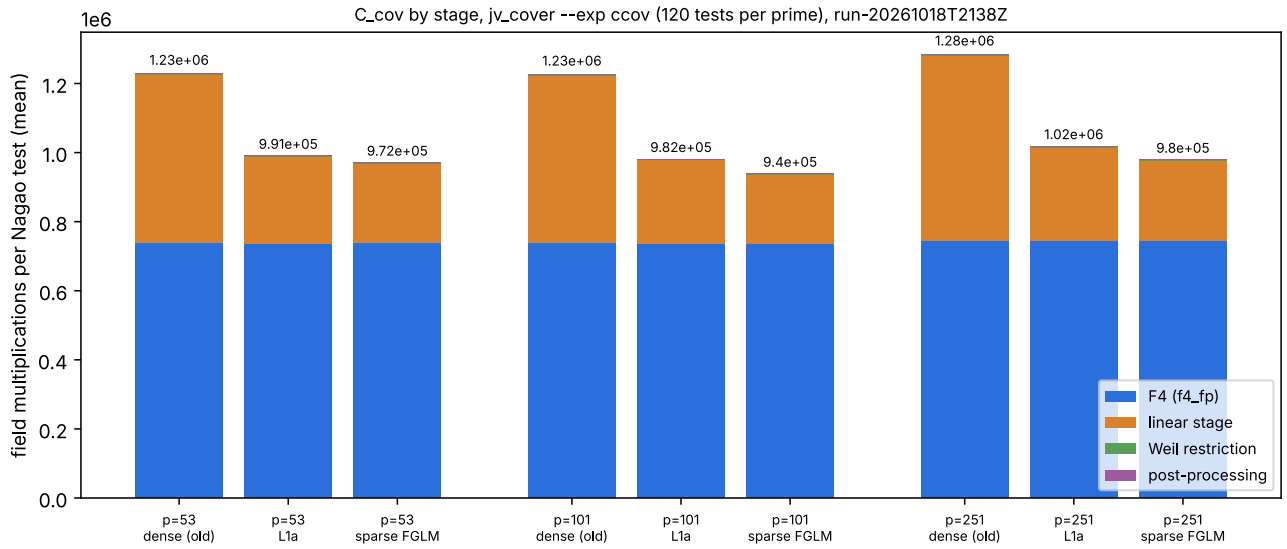


Figure 1: C_{cov} by stage, three arms.

p	planted	lin dense	lin sparse	lin L1a	C_{cov} dense	C_{cov} sparse (ratio)	L1a (ratio)
53	20/20 x3	4.875e5	2.296e5	2.490e5	1.2296e6	9.718e5 (1.265)	9.912e5 (1.241)
101	20/20 x3	4.863e5	1.991e5	2.409e5	1.2274e6	9.403e5 (1.305)	9.820e5 (1.250)
251	20/20 x3	5.347e5	2.304e5	2.690e5	1.2844e6	9.800e5 (1.311)	1.0186e6 (1.261)

Gates, every arm: planted 20/20, mismatches 0, unverified 0, incomplete 0. Dense `nonseparating` 3, 2, 0 (a random ℓ failed); sparse falls back to dense in 4, 0, 1 of 120 tests (p = 53, 101, 251), L1a likewise; the failed attempt is paid in the sparse cost. Linear stage: 2.12x, 2.44x, 2.32x cheaper.

End to end (`dlp`)

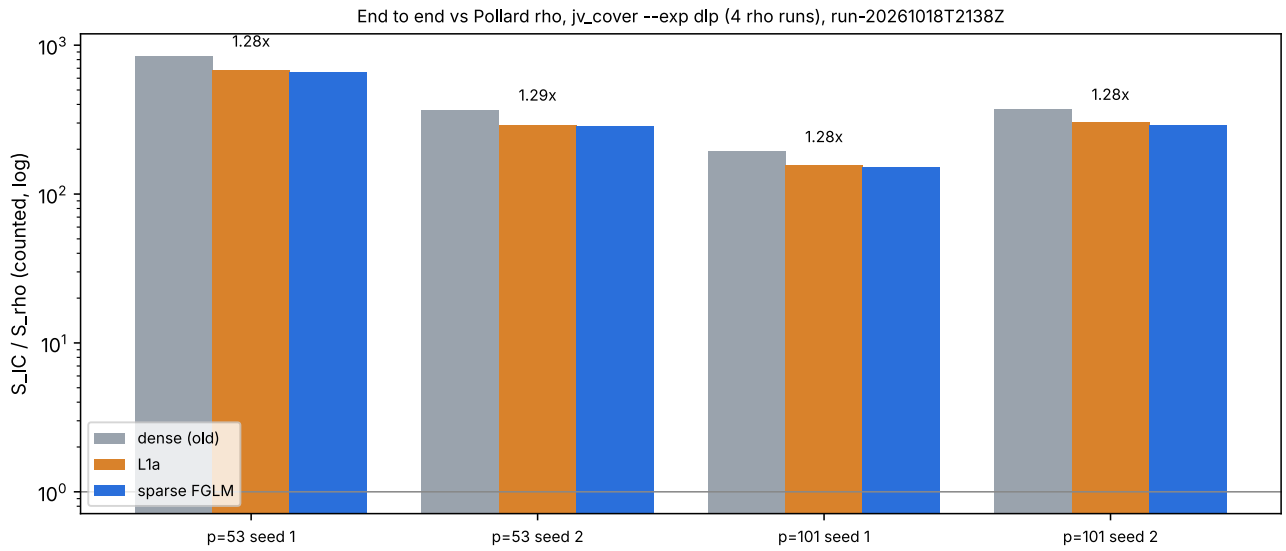


Figure 2: S/S_{rho} by arm.

p, seed	relations	correct	S dense	S sparse	ratio	S/S_{rho} dense	S/S_{rho} sparse (L1a)
53, 1	31	yes	937.2	731.6	1.281	851	664 (683)
53, 2	33	yes	424.5	328.8	1.291	369	286 (293)
101, 1	53	yes	287.8	224.4	1.282	194	151 (157)
101, 2	50	yes	466.3	365.5	1.276	373	293 (304)

Residual counts and relation counts are identical in all three arms; every logarithm is solved and correct. S_{rho} means (4 runs): 1.10, 1.15, 1.48, 1.25.

Verdict

- Estimate (210–260k linear, 1.22–1.29x) **beaten**: measured 199–230k linear and 1.27–1.31x C_{cov} (measured). The cheaper L1a variant reaches 241–269k and 1.24–1.26x.
- End to end S improves 1.28x; distance to rho falls from 194–851x to 151–664x. Relation collection stays over 99.9 % of S.
- Sparse beats L1a by 15 % of the linear stage: Wiedemann on the sparse M_{x_6} is cheaper than its Hessenberg form.
- Limits: counted cost only (wall time is observation on a shared host); unpriced native work is not emitted, so S/S_{rho} is a lower-bound comparison; 1 seed per prime for ccov, 2 per prime for dlp. Retries are 4 Wiedemann draws; a non-separating x_6 (1–3 % of tests) wastes them, so 2 draws would cut the linear stage a little more (proposed, not measured).
- Two existing tests, `stopped_f4_agrees_with_the_full_solver_and_the_oracle` and `replayed_f4_agrees_with_the_full_solver_and_the_oracle`, fail (staircase 29 vs 30, delta 62 vs 64) under every `JV_LIN` arm, i.e. on F4-level assertions; not verified whether they pass on unmodified main (the single rng draw shifts their random residuals).